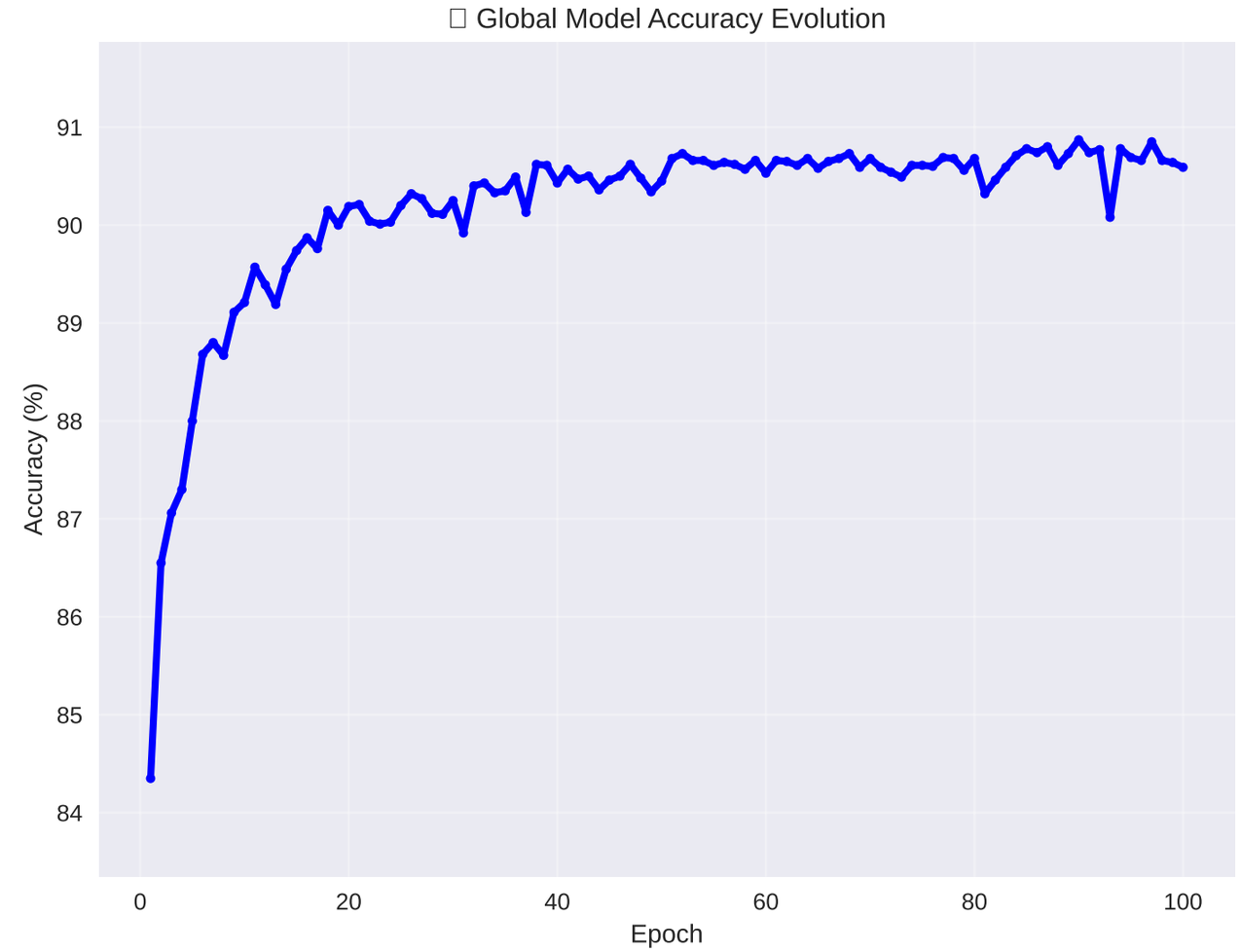
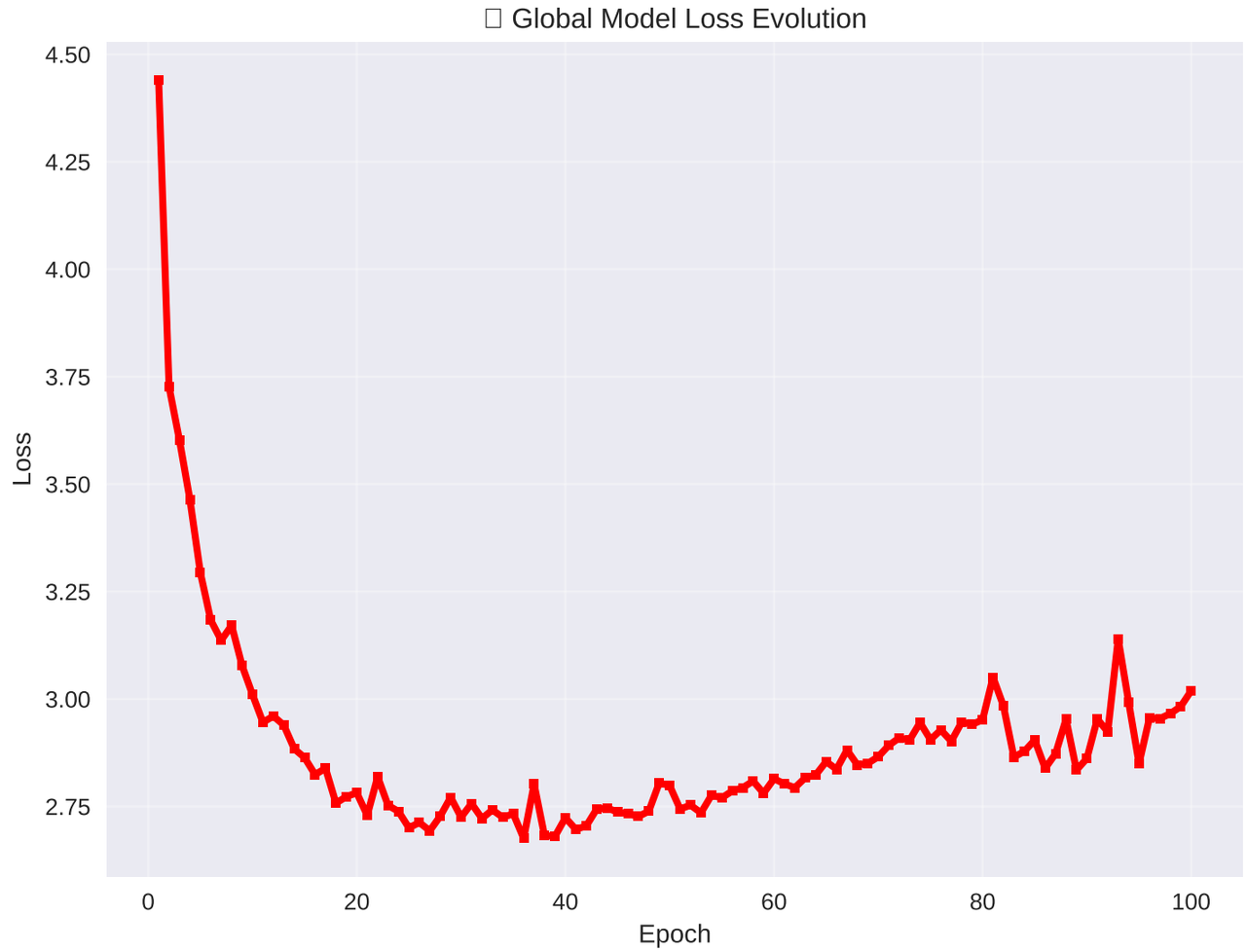
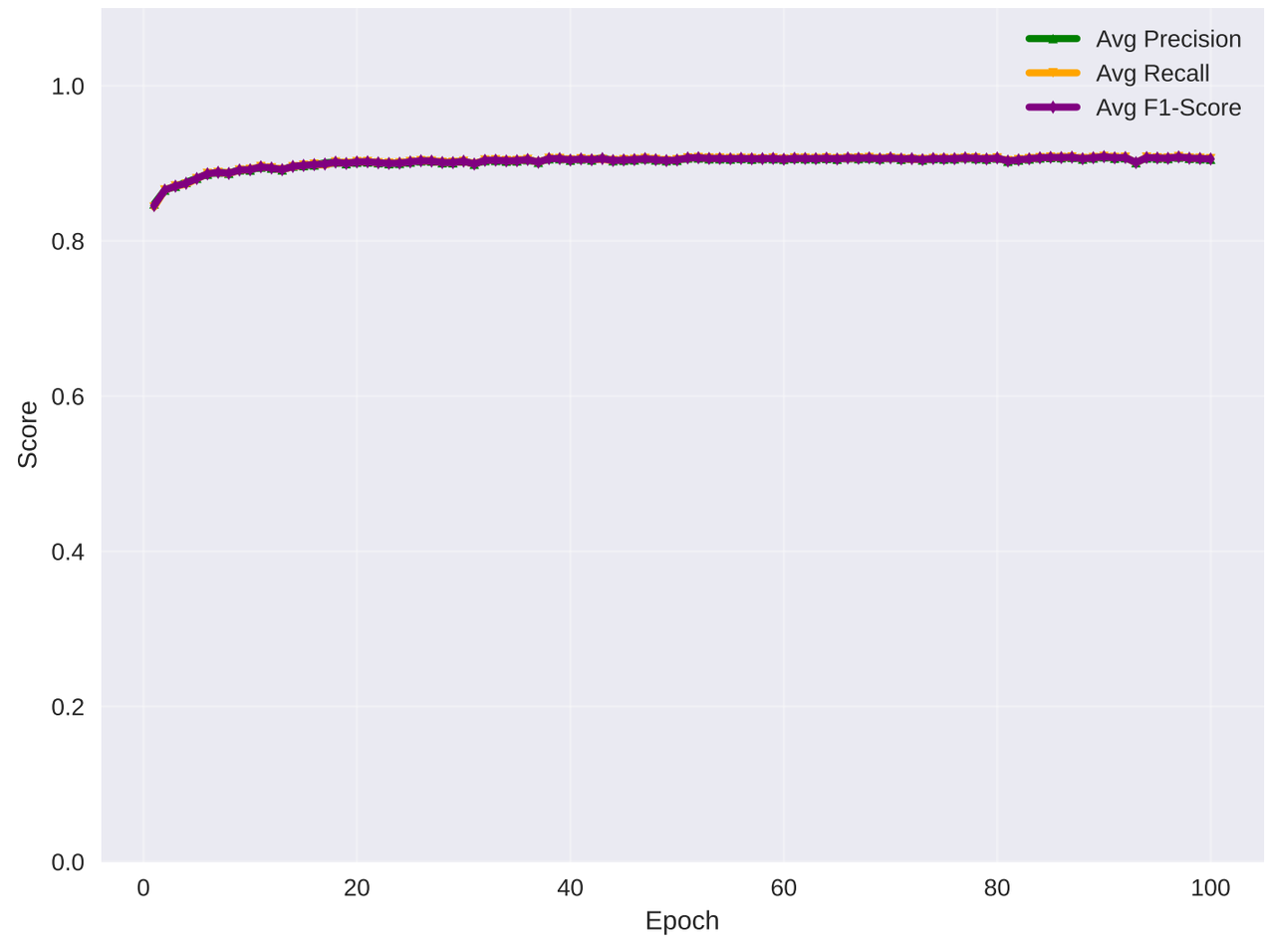
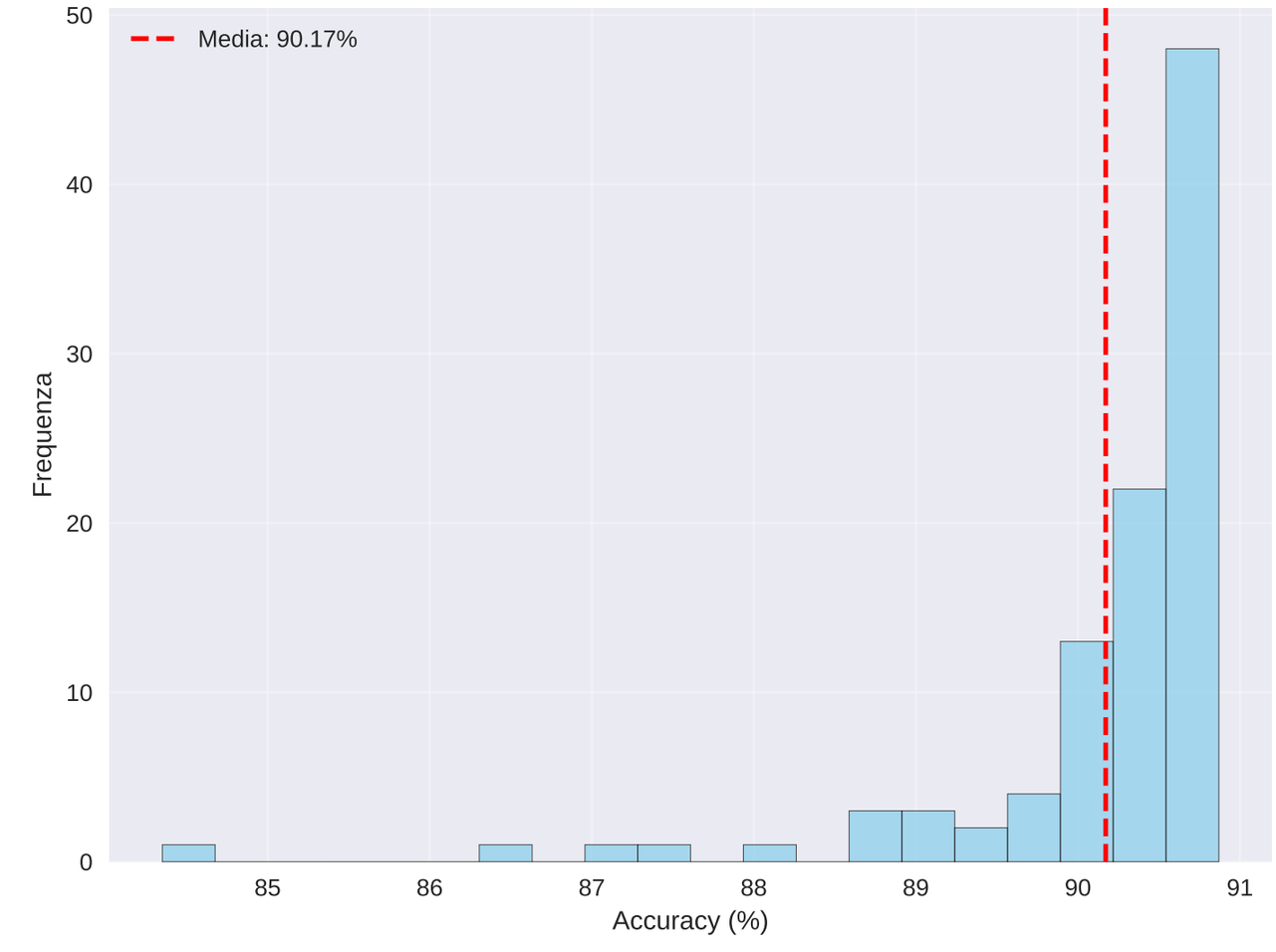
# Federated Learning - Main Metrics Overview

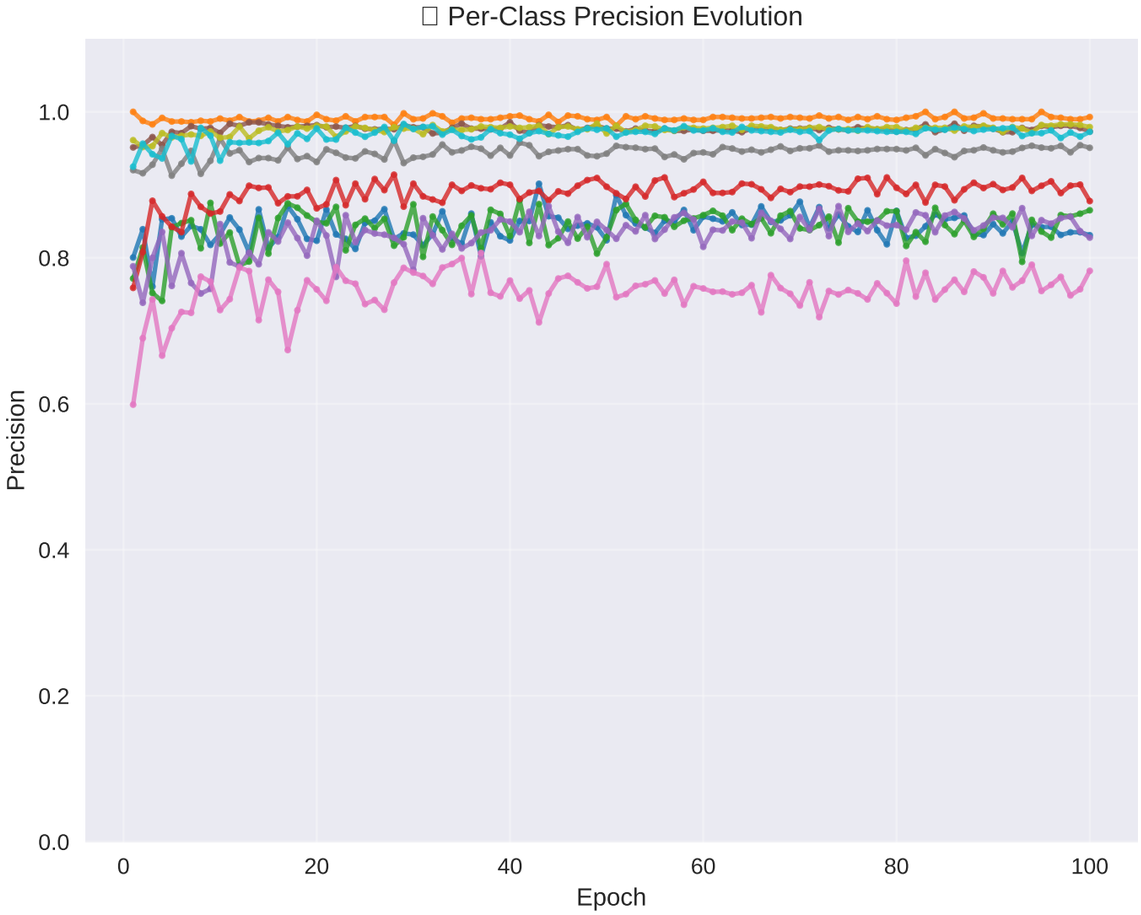


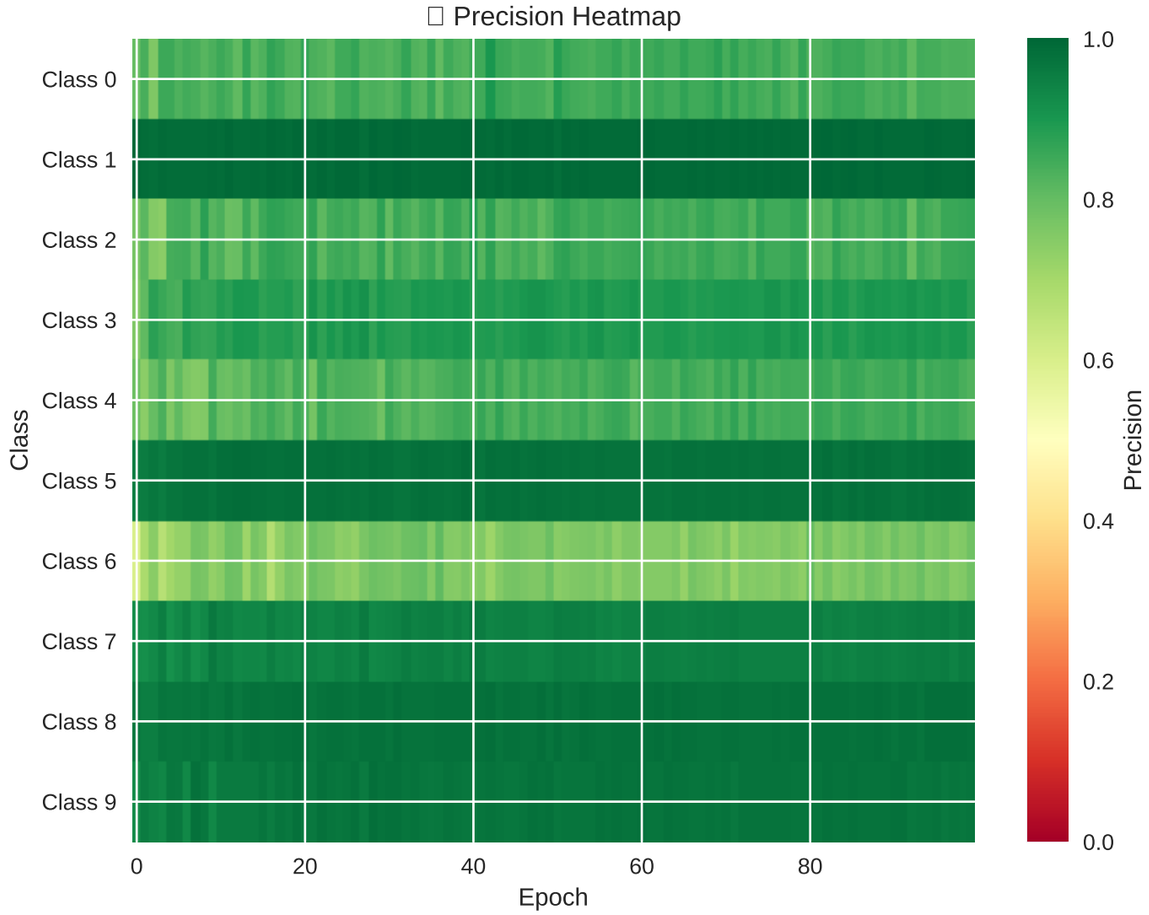



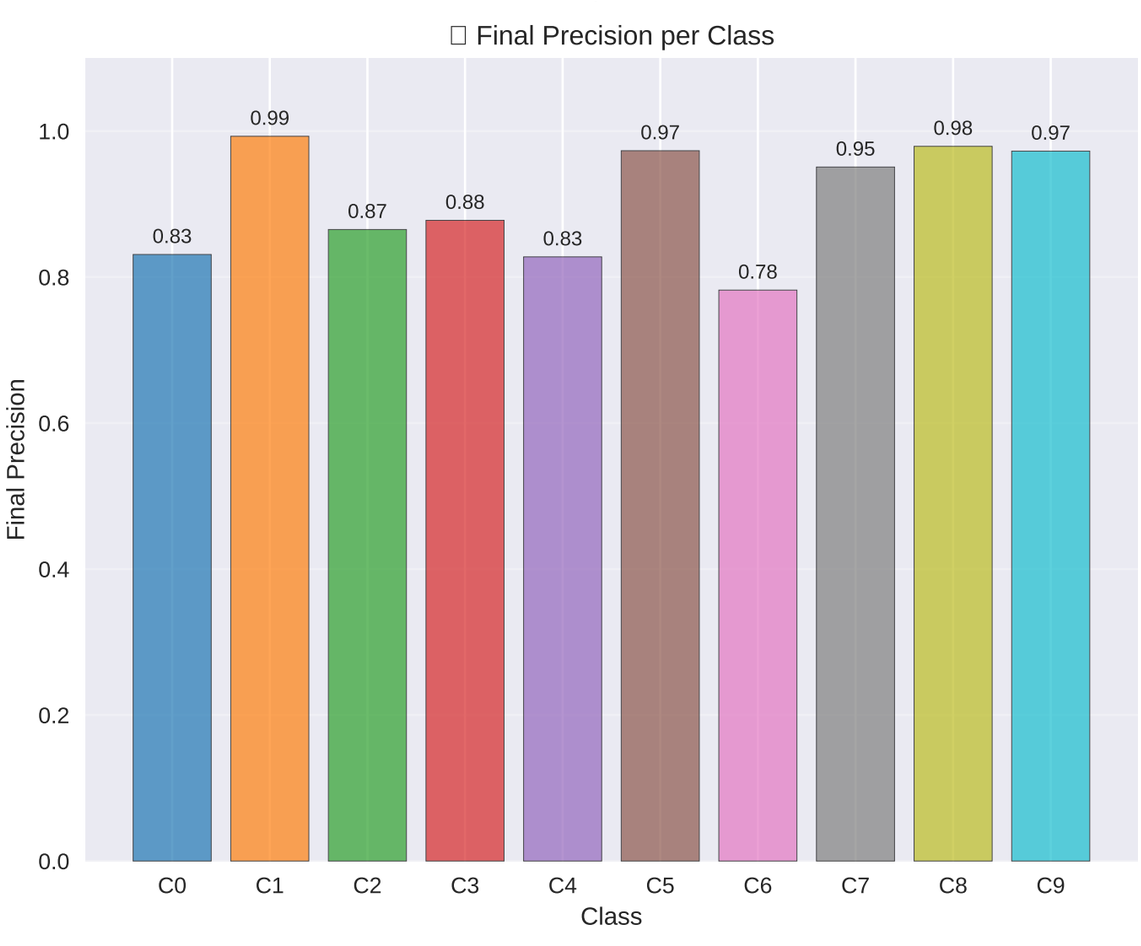



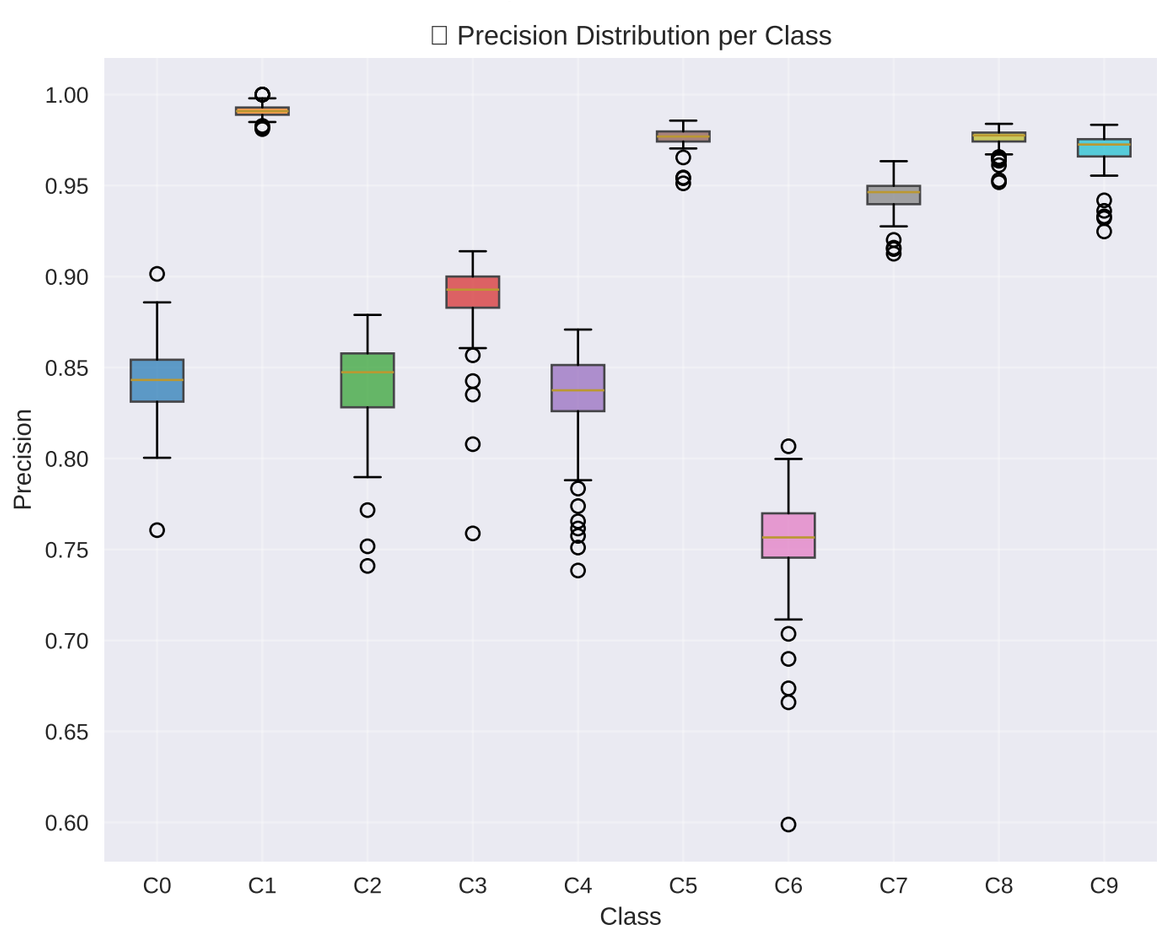



# Per-Class Precision Analysis



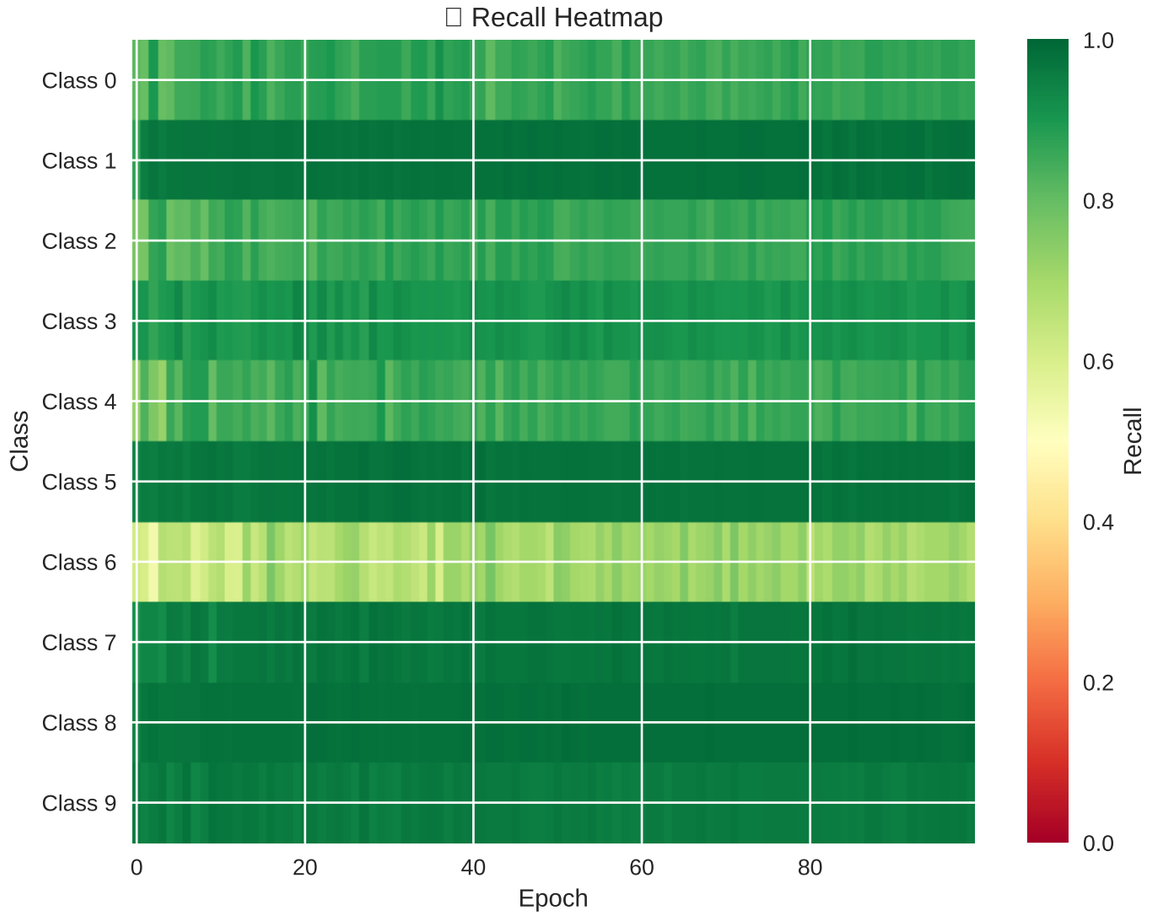



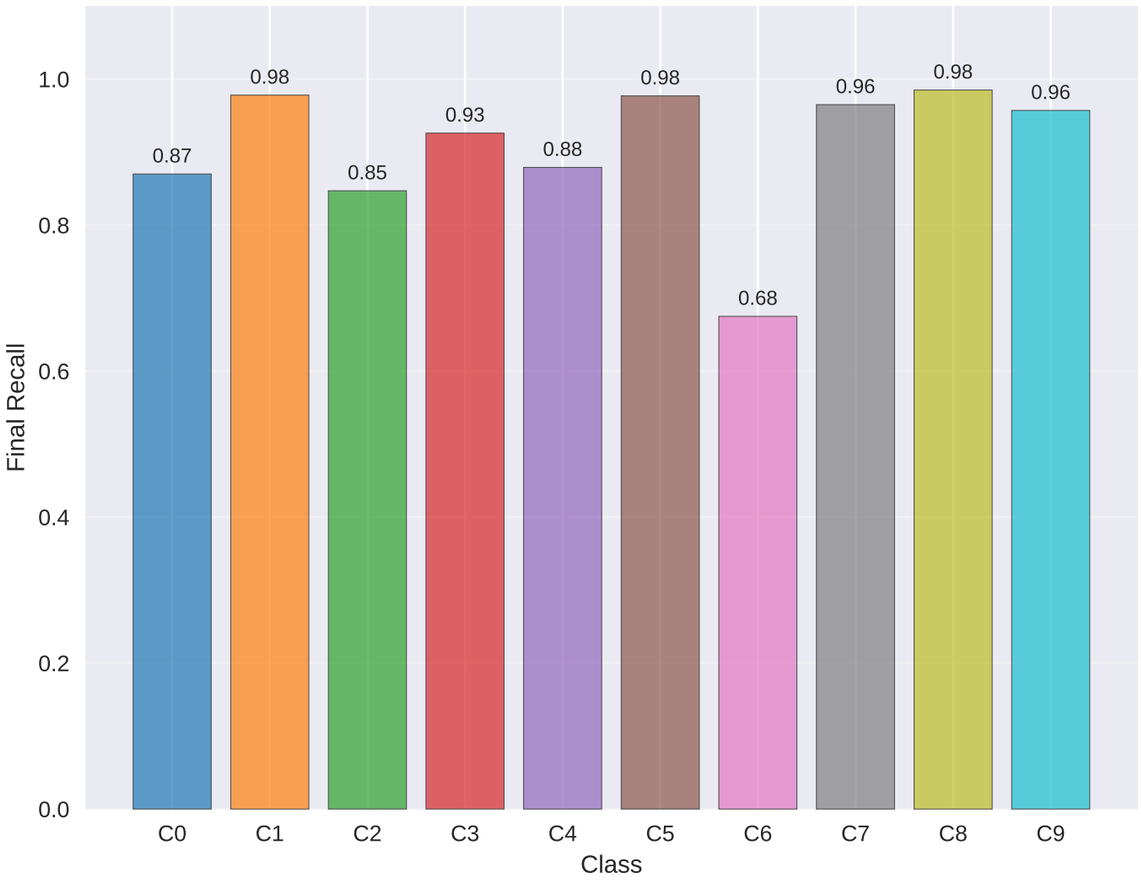



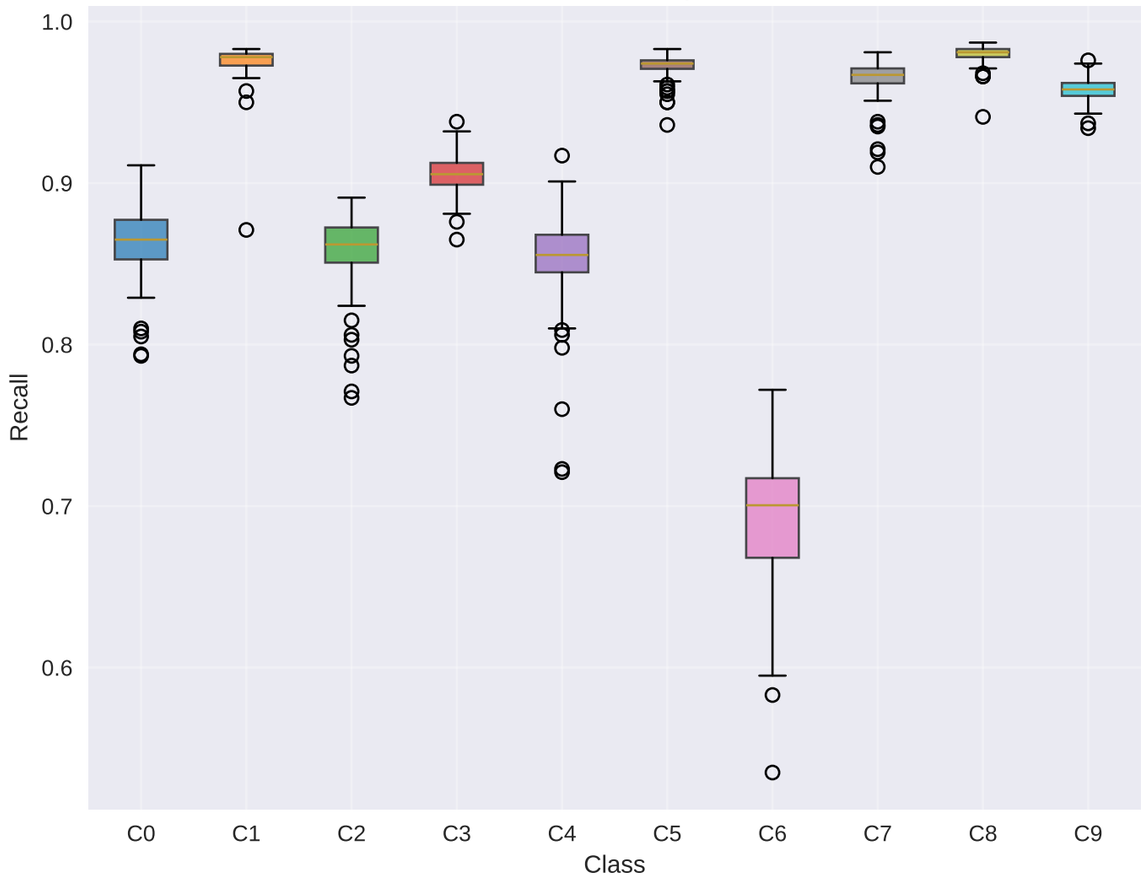



# Per-Class Recall Analysis

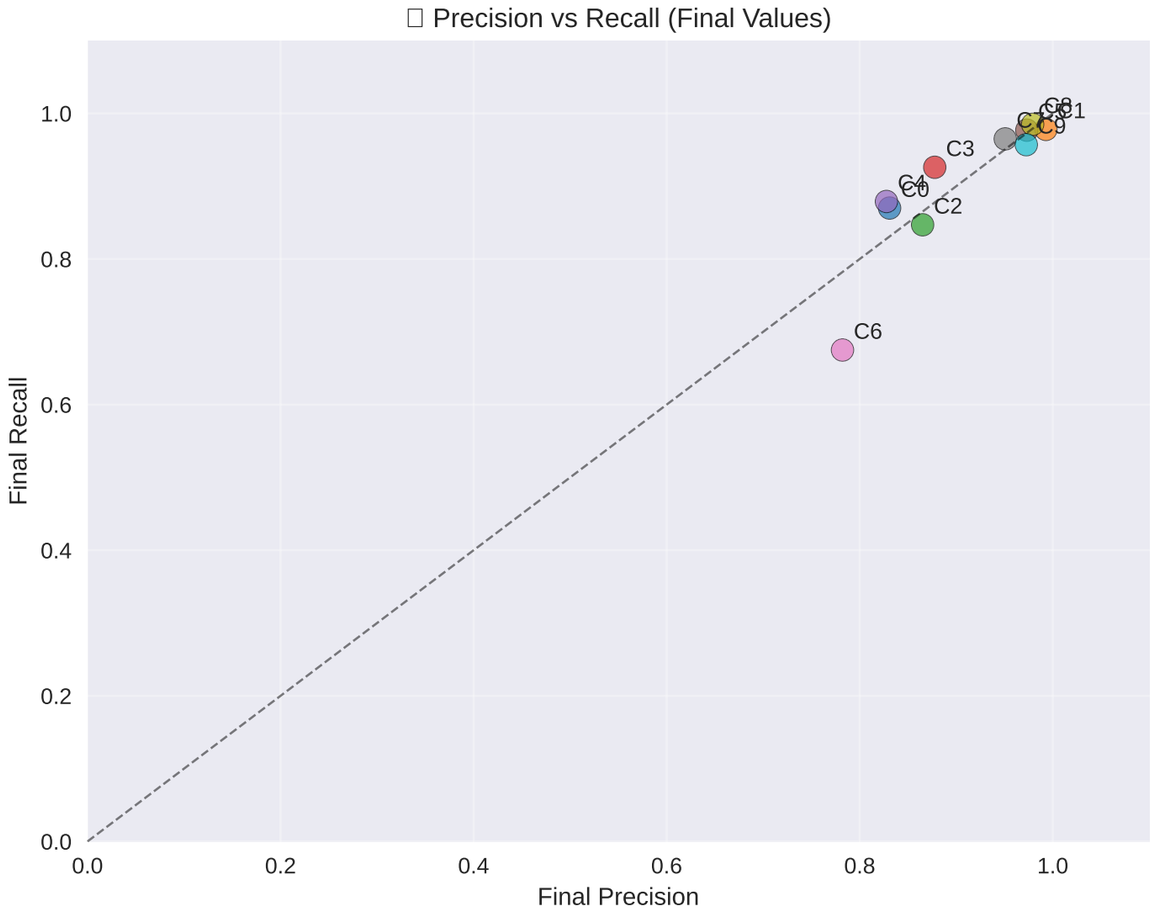


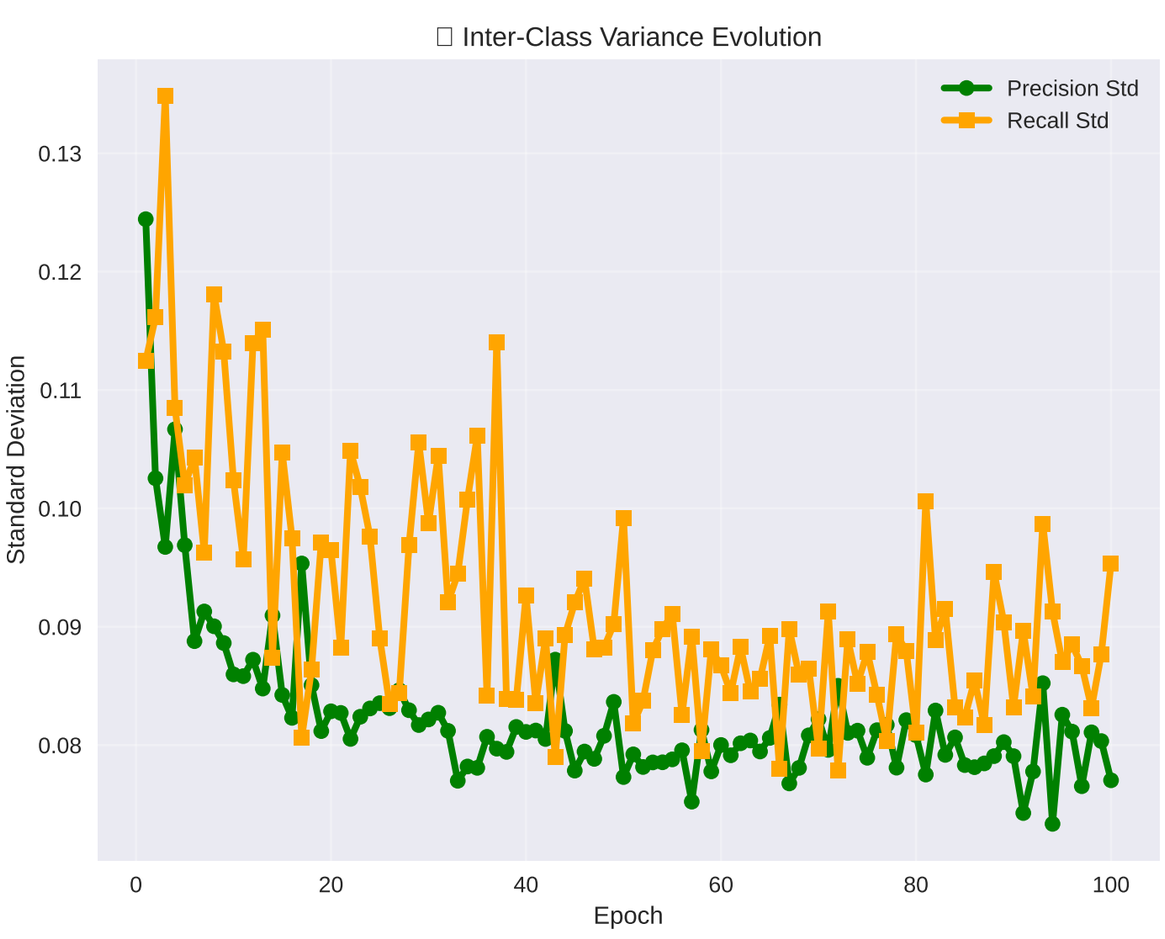



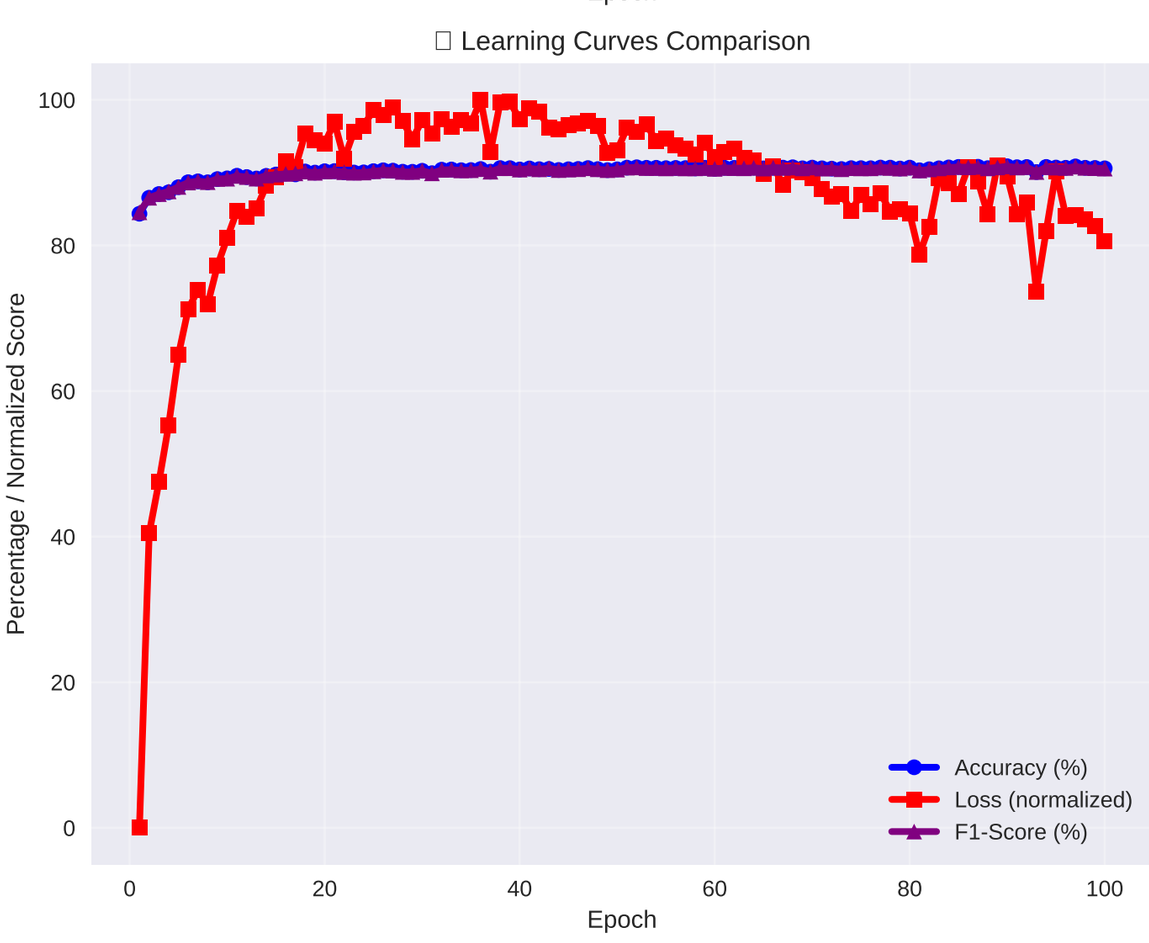
# Comparative Analysis









# Performance Summary Report

```
FEDERATED LEARNING - PERFORMANCE SUMMARY
========================================

FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-01 18:27:52
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 90.59%
• Loss: 3.0193
• Average Precision: 0.905
• Average Recall: 0.906
• Average F1-Score: 0.906

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.24%
• Riduzione Loss: +1.4213
• Stabilità Accuracy (CV): 0.0108

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.993)
• Peggiore Precision: Classe 6 (0.782)
• Migliore Recall: Classe 8 (0.985)
• Peggiore Recall: Classe 6 (0.675)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.831, R=0.870, F1=0.850
• Classe 1: P=0.993, R=0.978, F1=0.985
• Classe 2: P=0.865, R=0.847, F1=0.856
• Classe 3: P=0.878, R=0.926, F1=0.901
• Classe 4: P=0.828, R=0.879, F1=0.853
• Classe 5: P=0.973, R=0.977, F1=0.975
• Classe 6: P=0.782, R=0.675, F1=0.725
• Classe 7: P=0.951, R=0.965, F1=0.958
• Classe 8: P=0.979, R=0.985, F1=0.982
• Classe 9: P=0.973, R=0.957, F1=0.965

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
```